Name:

Quiz 3 Solutions

Instructions. The only tools you are permitted to use are pencils, pens, erasers, a sheet of handwritten notes, and your mind. No electronic devices! You have 1 hour. Be sure to sign the Honor Code statement on the last page. Good luck!

Problem 1 (True-false, 6 points)**.** You will get 1 point for a correct answer and 0 points for blank and incorrect answers.[1] No explanations are required.

(1) If λ is a repeated eigenvalue of a 2×2 matrix, it must be defective. T **F**

(2) For any real number a, one of the eigenvalues of the matrix $\begin{bmatrix} 1 & a \\ a & 1 \end{bmatrix}$ must be a nonnegative real number. **T** F

(3) If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$, then $e^A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. **T** F

(4) If $\vec{x}\,' = A\vec{x}$ for $A = \begin{bmatrix} -1 & 1 & 3 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$, then it must be that $\lim_{t\to\infty} \vec{x}(t) = \vec{0}$. **T** F

(5) If A is a 3×3 matrix and $\lambda = 3$ is an eigenvalue of A with geometric multiplicity 3, then it must be that $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. **T** F

(6) If a $n \times n$ matrix of functions X is a fundamental matrix solution for a homogeneous linear system $\vec{x}\,' = A\vec{x}$, then XP must also be a fundamental matrix solution for any matrix of constants P. T **F**

[1] During quiz revisions, you'll have the chance to meet with me one-on-one and convince me that you understand and can fully explain *up to 2 of the true-false questions that you left blank* to get full credit on those questions.

Problem 2 (2 points)**.** Calculate the general solution of the ODE $x'' - 8x' + 16x = 0$.

Solution. We guess $x(t) = e^{rt}$ and find that $r^2 - 8r + 16 = 0$, which means that $r = 4$ is a repeated root of the characteristic equation. Thus e^{4t} and te^{4t} are both solutions, and the general solution is of the form

$$x(t) = C_1 e^{4t} + C_2 t e^{4t}$$

for real constants C_1 and C_2.

More work, but ultimately equivalent, is to set up the matrix system

$$\frac{d}{dt}\begin{bmatrix} x \\ x' \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -16 & 8 \end{bmatrix}}_{A} \begin{bmatrix} x \\ x' \end{bmatrix}$$

and then use the eigenvalue method to solve. This matrix A will have a deficient eigenvalue at $\lambda = 4$.

Problem 3 (4 points)**.** In each of the following, a matrix A is given, together with some information about its eigenvalues. Use the given information to write down a real-valued fundamental matrix solution for the system $\vec{x}\,' = A\vec{x}$.

(a) $A = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$ has eigenvalues -1 and 1, and corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$, respectively.

Solution. The functions $e^{-t}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $e^{t}\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ form a basis for the solution space, so a fundamental matrix solution is

$$X = \begin{bmatrix} e^{-t} & 3e^{t} \\ e^{-t} & e^{t} \end{bmatrix}.$$

(b) The matrix $\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ has eigenvalues $2 \pm i$, and an eigenvector corresponding to $\lambda = 2 + i$ is $\begin{bmatrix} 1-i \\ 1 \end{bmatrix}$.

Solution. One complex-valued solution to the system is given by

$$\begin{aligned} e^{(2+i)t}\begin{bmatrix} 1-i \\ i \end{bmatrix} &= e^{2t}(\cos(t) + i\sin(t))\begin{bmatrix} 1-i \\ 1 \end{bmatrix} \\ &= e^{2t}\begin{bmatrix} \cos(t) - \sin(t) + i(\sin(t) - \cos(t)) \\ \cos(t) + i\sin(t) \end{bmatrix} \\ &= e^{2t}\begin{bmatrix} \cos(t) + \sin(t) \\ \cos(t) \end{bmatrix} + ie^{2t}\begin{bmatrix} \sin(t) - \cos(t) \\ \sin(t) \end{bmatrix}. \end{aligned}$$

Taking real and imaginary parts, we find that

$$X = e^{2t}\begin{bmatrix} \cos(t) + \sin(t) & \sin(t) - \cos(t) \\ \cos(t) & \sin(t) \end{bmatrix}.$$

(c) $A = \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$ has defective eigenvalue 3, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector, and $(A - 2I)\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Solution. The functions $e^{3t}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $e^{3t}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} + t\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$ form a basis for the solution space. Thus

$$X = \begin{bmatrix} e^{3t} & 2e^{3t} + te^{3t} \\ e^{3t} & te^{3t} \end{bmatrix} = e^{3t}\begin{bmatrix} 1 & 2+t \\ 1 & t \end{bmatrix}.$$

(d) $A = \begin{bmatrix} -1 & 7 & -8 \\ -1 & 3 & 0 \\ 0 & -1 & 4 \end{bmatrix}$ has eigenvalue 2 and $\vec{v}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ is an eigenvector. Moreover, $\vec{v}_2 = \begin{bmatrix} -3 \\ -1 \\ 0 \end{bmatrix}$ and $\vec{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ are vectors such that $(A - 2I)\vec{v}_2 = \vec{v}_1$ and $(A - 2I)\vec{v}_3 = \vec{v}_2$.

Solution. The solution space is spanned by the following three functions.

$$\vec{x}_1(t) = e^{2t}\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \qquad \vec{x}_2(t) = e^{2t}\left(\begin{bmatrix} -3 \\ -1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}\right) \qquad \vec{x}_3(t) = e^{2t}\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -3 \\ -1 \\ 0 \end{bmatrix} + \frac{t^2}{2}\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}\right)$$

Thus

$$X = e^{2t}\begin{bmatrix} 2 & -3+2t & 1-3t+t^2 \\ 2 & -1+2t & -t+t^2 \\ 1 & t & t^2/2 \end{bmatrix}$$

is a fundamental matrix solution.

Problem 4 (3 points)**.** For each of the following matrices A, compute the matrix exponential e^A. If you like, you can express your answer as a product of matrices; you do not need to multiply them out.

(a) $A = \begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix}$, which has the property that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Solution. This matrix is diagonalizable, since $(1,1)$ is an eigenvector with eigenvalue -1 and $(2,1)$ is an eigenvector with eigenvalue $+1$. We have

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}^{-1}$$

which means that

$$e^A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^{-1} & 0 \\ 0 & e \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 2e - e^{-1} & 2e^{-1} - 2e \\ e - e^{-1} & 2e^{-1} - e \end{bmatrix}.$$

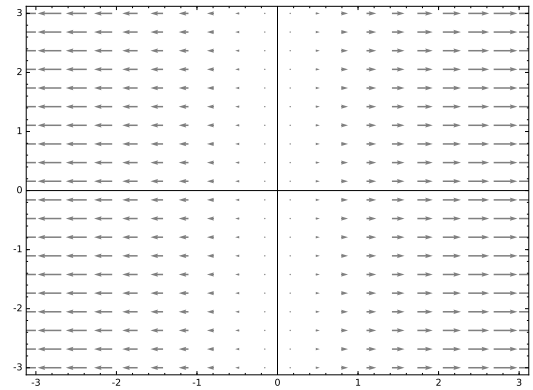
(b) $A = \begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$.

Solution. Observe that $S = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ and $N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ commute, S is diagonal, and N is nilpotent with $N^2 = 0$. Thus $e^N = I + N$ and

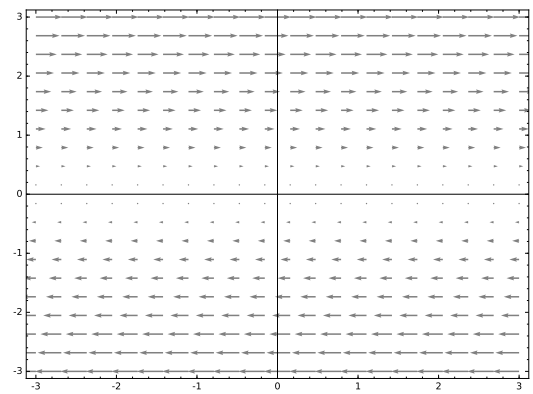
$$e^A = e^{S+N} = e^S e^N = \begin{bmatrix} e^{-1} & 0 \\ 0 & e^{-1} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^{-1} & 2e^{-1} \\ 0 & e^{-1} \end{bmatrix}.$$

Problem 5 (3 points)**.** On the left are descriptions of several 2×2 matrices. On the right are several phase portraits. Match each matrix A with the phase portrait of the system $\vec{x}\,' = A\vec{x}$.

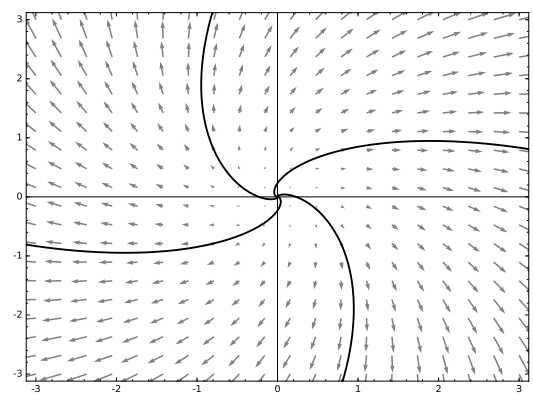
(a) A has eigenvalues $2 \pm i$.



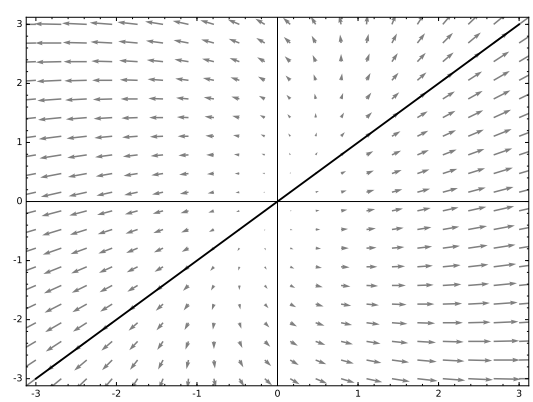
(b) $\lambda = 1$ is a deficient eigenvalue of A.



(c) $\lambda = 0$ is a deficient eigenvalue of A.



(d) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



Solution. The phase portraits belong to the systems (d), (c), (a), (b), respectively.

Problem 6 (Extra credit! 2 points, no partial credit)**.** Consider the system $\vec{x}' = A\vec{x}$ where

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

Let $x = x_1 + x_2 + x_3 + x_4 + x_5$ and suppose further that $x(0) = 100$. What is $\lim_{t\to\infty} x(t)$?

Solution. No, this problem is most definitely not asking you to apply the eigenvalue method to a 5×5 matrix. There's a trick; can you find it? :)

Honor code. If you have neither given nor received any unauthorized aid on this quiz, please write either "HCU" or "Honor Code Upheld" below, and sign your name next to it.

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